



SE3082 – lab 07

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Lab_07
supundharmaratne@Supuns-MacBook-Air Lab_07 % ssh -i ~/Downloads/Lab07Key.pem ec2-user@54.211.145.38
Rank 7: summed local chunk of size 125000 => local_sum = 117187562500

[Gather] Total sum = 500000500000
[Gather] Expected = 500000500000
[Gather] Correct? = YES
[Gather] Time = 1.3800 sec

=== Exercise 4: Scatter + Reduce (np=8) ===
mpirun -np 8 ./Exercise04/sum_reduce
Root filled array with values 1 to 1000000
Rank 0: summed local chunk of size 125000 => local_sum = 7812562500
Rank 1: summed local chunk of size 125000 => local_sum = 23437562500
Rank 2: summed local chunk of size 125000 => local_sum = 39062562500
Rank 4: summed local chunk of size 125000 => local_sum = 70312562500
Rank 3: summed local chunk of size 125000 => local_sum = 54687562500
Rank 6: summed local chunk of size 125000 => local_sum = 101562562500
Rank 5: summed local chunk of size 125000 => local_sum = 85937562500
Rank 7: summed local chunk of size 125000 => local_sum = 117187562500

[Reduce] Total sum = 500000500000
[Reduce] Expected = 500000500000
[Reduce] Correct? = YES
[Reduce] Time = 1.3600 sec

=== Exercise 5: Scatter + Allreduce (np=8) ===
mpirun -np 8 ./Exercise05/sum_allreduce
Root filled array with values 1 to 1000000
Rank 2: local_sum = 39062562500, total_sum = 500000500000, contribution = 7.8125%
Rank 3: local_sum = 54687562500, total_sum = 500000500000, contribution = 10.9375%
Rank 4: local_sum = 70312562500, total_sum = 500000500000, contribution = 14.0625%
Rank 5: local_sum = 85937562500, total_sum = 500000500000, contribution = 17.1875%
Rank 7: local_sum = 117187562500, total_sum = 500000500000, contribution = 23.4375%
Rank 6: local_sum = 101562562500, total_sum = 500000500000, contribution = 20.3125%
Rank 1: local_sum = 23437562500, total_sum = 500000500000, contribution = 4.6875%
Rank 0: local_sum = 7812562500, total_sum = 500000500000, contribution = 1.5625%

[Allreduce] Total sum = 500000500000
[Allreduce] Expected = 500000500000
[Allreduce] Correct? = YES
[Allreduce] Time = 1.4303 sec

=== Exercise 6: Scatter + Scan (np=8) ===
mpirun -np 8 ./Exercise06/sum_scan
Root filled array with values 1 to 1000000
Rank 0: local_sum = 7812562500, prefix_sum = 7812562500, sum_before_me = 0 (expected_prefix = 7812562500, match = YES)
Rank 1: local_sum = 23437562500, prefix_sum = 31250125000, sum_before_me = 7812562500 (expected_prefix = 31250125000, match = YES)
Rank 4: local_sum = 70312562500, prefix_sum = 195312812500, sum_before_me = 125000250000 (expected_prefix = 195312812500, match = YES)
Rank 6: local_sum = 101562562500, prefix_sum = 382812937500, sum_before_me = 281250375000 (expected_prefix = 382812937500, match = YES)
Rank 7: local_sum = 117187562500, prefix_sum = 500000500000, sum_before_me = 382812937500 (expected_prefix = 500000500000, match = YES)

[Scan] Final prefix_sum (should equal global total) = 500000500000
[Scan] Expected = 500000500000
[Scan] Correct? = YES
[Scan] Time = 1.5173 sec
Rank 5: local_sum = 85937562500, prefix_sum = 281250375000, sum_before_me = 195312812500 (expected_prefix = 281250375000, match = YES)
Rank 2: local_sum = 39062562500, prefix_sum = 70312687500, sum_before_me = 31250125000 (expected_prefix = 70312687500, match = YES)
Rank 3: local_sum = 54687562500, prefix_sum = 125000250000, sum_before_me = 70312687500 (expected_prefix = 125000250000, match = YES)

[ec2-user@ip-172-31-80-182 Lab_07]$

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Lab_07
supundharmaratne@Supuns-MacBook-Air Lab_07 % ssh -i ~/Downloads/Lab07Key.pem ec2-user@54.211.145.38
=== Exercise 3: Scatter + Gather (np=2) ===
mpirun -np 2 ./Exercise03/sum_gather
Root filled array with values 1 to 1000000
Rank 0: summed local chunk of size 500000 => local_sum = 125000250000
Rank 1: summed local chunk of size 500000 => local_sum = 375000250000

[Gather] Total sum = 500000500000
[Gather] Expected = 500000500000
[Gather] Correct? = YES
[Gather] Time = 0.1598 sec

=== Exercise 4: Scatter + Reduce (np=2) ===
mpirun -np 2 ./Exercise04/sum_reduce
Root filled array with values 1 to 1000000
Rank 0: summed local chunk of size 500000 => local_sum = 125000250000
Rank 1: summed local chunk of size 500000 => local_sum = 375000250000

[Reduce] Total sum = 500000500000
[Reduce] Expected = 500000500000
[Reduce] Correct? = YES
[Reduce] Time = 0.1635 sec

=== Exercise 5: Scatter + Allreduce (np=2) ===
mpirun -np 2 ./Exercise05/sum_allreduce
Root filled array with values 1 to 1000000
Rank 1: local_sum = 375000250000, total_sum = 500000500000, contribution = 75.0000%
Rank 0: local_sum = 125000250000, total_sum = 500000500000, contribution = 25.0000%

[Allreduce] Total sum = 500000500000
[Allreduce] Expected = 500000500000
[Allreduce] Correct? = YES
[Allreduce] Time = 0.1700 sec

=== Exercise 6: Scatter + Scan (np=2) ===
mpirun -np 2 ./Exercise06/sum_scan
Root filled array with values 1 to 1000000
Rank 0: local_sum = 125000250000, prefix_sum = 125000250000, sum_before_me = 0 (expected_prefix = 125000250000, match = YES)
Rank 1: local_sum = 375000250000, prefix_sum = 500000500000, sum_before_me = 125000250000 (expected_prefix = 500000500000, match = YES)

[Scan] Final prefix_sum (should equal global total) = 500000500000
[Scan] Expected = 500000500000
[Scan] Correct? = YES
[Scan] Time = 0.1572 sec

[ec2-user@ip-172-31-80-182 Lab_07]$

Lab_07
supundharmaratne@Supuns-MacBook-Air Lab_07 % ssh -i ~/Downloads/Lab07Key.pem ec2-user@54.211.145.38
[ec2-user@ip-172-31-80-182 Lab_07]$ make run-reduce
=== Exercise 4: Scatter + Reduce (np=4) ===
mpirun -np 4 ./Exercise04/sum_reduce
Root filled array with values 1 to 1000000
Rank 0: summed local chunk of size 250000 => local_sum = 31250125000
Rank 1: summed local chunk of size 250000 => local_sum = 93750125000
Rank 2: summed local chunk of size 250000 => local_sum = 156250125000
Rank 3: summed local chunk of size 250000 => local_sum = 218750125000

[Reduce] Total sum = 500000500000
[Reduce] Expected = 500000500000
[Reduce] Correct? = YES
[Reduce] Time = 0.5151 sec

[ec2-user@ip-172-31-80-182 Lab_07]$ make run-allreduce
=== Exercise 5: Scatter + Allreduce (np=4) ===
mpirun -np 4 ./Exercise05/sum_allreduce
Root filled array with values 1 to 1000000
Rank 3: local_sum = 218750125000, total_sum = 500000500000, contribution = 43.7500%
Rank 1: local_sum = 93750125000, total_sum = 500000500000, contribution = 18.7500%
Rank 2: local_sum = 156250125000, total_sum = 500000500000, contribution = 31.2500%
Rank 0: local_sum = 31250125000, total_sum = 500000500000, contribution = 6.2500%

[Allreduce] Total sum = 500000500000
[Allreduce] Expected = 500000500000
[Allreduce] Correct? = YES
[Allreduce] Time = 0.5401 sec

[ec2-user@ip-172-31-80-182 Lab_07]$ make run-scan
=== Exercise 6: Scatter + Scan (np=4) ===
mpirun -np 4 ./Exercise06/sum_scan
Root filled array with values 1 to 1000000
Rank 0: local_sum = 31250125000, prefix_sum = 31250125000, sum_before_me = 0 (expected_prefix = 31250125000, match = YES)
Rank 1: local_sum = 93750125000, prefix_sum = 125000250000, sum_before_me = 31250125000 (expected_prefix = 125000250000, match = YES)
Rank 2: local_sum = 156250125000, prefix_sum = 218750375000, sum_before_me = 125000250000 (expected_prefix = 218750375000, match = YES)
Rank 3: local_sum = 218750125000, prefix_sum = 500000500000, sum_before_me = 218750375000 (expected_prefix = 500000500000, match = YES)

[Scan] Final prefix_sum (should equal global total) = 500000500000
[Scan] Expected = 500000500000
[Scan] Correct? = YES
[Scan] Time = 0.4675 sec

[ec2-user@ip-172-31-80-182 Lab_07]$

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The image displays two screenshots of a VS Code terminal window, labeled 'Lab_07', showing the execution of MPI exercises.

Top Screenshot: The terminal shows the execution of 'make run-scatter'. The output indicates that the root filled an array with values 1 to 1000000, and four ranks (0-3) summed local chunks of size 250000. The total sum is 500000500000, and the exercise is correct.

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supundharmaratne@Supuns-MacBook-Air Lab_07 % ssh -i ~/Downloads/Lab07Key.pem ec2-user@54.211.145.38
[ec2-user@ip-172-31-80-182 Lab_07]$
[ec2-user@ip-172-31-80-182 Lab_07]$ make run-scatter
=== Exercise 2: Scatter + Send/Recv (np=4) ===
mpirun -np 4 ./Exercise02/sum_scatter
Root filled array with values 1 to 1000000
Rank 0: summed local chunk of size 250000 => local_sum = 31250125000
Rank 1: summed local chunk of size 250000 => local_sum = 93750125000
Rank 2: summed local chunk of size 250000 => local_sum = 156250125000
Rank 3: summed local chunk of size 250000 => local_sum = 218750125000

[Scatter] Total sum = 500000500000
[Scatter] Expected = 500000500000
[Scatter] Correct? = YES
[Scatter] Time = 0.5258 sec
[ec2-user@ip-172-31-80-182 Lab_07]$

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Bottom Screenshot: The terminal shows the execution of 'make run-bcast'. It fails with an error: 'make: *** No rule to make target 'run-bcast'. Stop.' and 'make: mpirun: No such file or directory'. After setting the PATH to include the mpich bin directory, the exercise runs successfully.

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supundharmaratne@Supuns-MacBook-Air Lab_07 % ssh -i ~/Downloads/Lab07Key.pem ec2-user@54.211.145.38
make: *** No rule to make target 'run-bcast'. Stop.
[ec2-user@ip-172-31-80-182 ~]$ cd Lab_07
make run-bcast
=== Exercise 1: Bcast + Send/Recv (np=4) ===
mpirun -np 4 ./Exercise01/sum_bcast
make: mpirun: No such file or directory
make: *** [Makefile:44: run-bcast] Error 127
[ec2-user@ip-172-31-80-182 Lab_07]$ export PATH=$PATH:/usr/lib64/mpich/bin
which mpirun
/usr/lib64/mpich/bin/mpirun
[ec2-user@ip-172-31-80-182 Lab_07]$ make run-bcast
=== Exercise 1: Bcast + Send/Recv (np=4) ===
mpirun -np 4 ./Exercise01/sum_bcast
Root filled array with values 1 to 1000000
Rank 0: summed indices [0, 250000) => local_sum = 31250125000
Rank 2: summed indices [500000, 750000) => local_sum = 156250125000
Rank 1: summed indices [250000, 500000) => local_sum = 93750125000
Rank 3: summed indices [750000, 1000000) => local_sum = 218750125000

[Bcast] Total sum = 500000500000
[Bcast] Expected = 500000500000
[Bcast] Correct? = YES
[Bcast] Time = 1.2901 sec
[ec2-user@ip-172-31-80-182 Lab_07]$ echo 'export PATH=$PATH:/usr/lib64/mpich/bin' >> ~/.bashrc
source ~/.bashrc
[ec2-user@ip-172-31-80-182 Lab_07]$

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